

Geometric Mahalanobis Cascade

A Novel Architecture for Gradient-Free Neural Networks

Breaking Through Plateau Limits by Interleaving
Evolved Projections with Geometric Class Measurement

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Context: CIFAR-10 Classification Research

Prior Result: 68.20% (v27 Ultimate Fusion)

1. Executive Summary

This document describes a novel architecture that addresses a fundamental limitation in gradient-free evolutionary neural networks: the plateau problem. Both evolved projection layers and Mahalanobis-based geometric classifiers individually reach performance ceilings when applied in isolation. This architecture breaks through these plateaus by interleaving the two approaches, creating a cascade where geometry informs evolution and evolution reshapes geometry.

Core Insight: Each system plateaus because it exhausts the information available in its fixed frame. Mahalanobis plateaus at 54% because it measures geometry in one projection. Freeze-and-stack plateaus because each layer evolves blind to class structure. But when interleaved, neither system ever sees the same data twice.

The architecture emerges from three years of GENREG research, building on the 68.20% CIFAR-10 accuracy achieved through evolved dendrite features and multi-source fusion. The Geometric Mahalanobis Cascade represents the next evolutionary step: moving from optimizing within fixed feature vocabularies to discovering the mathematical operations themselves.

2. The Plateau Problem

2.1 Mahalanobis Plateau

Mahalanobis distance classification achieves 54.24% accuracy on geometric shape features by measuring each sample's distance to class centroids using per-class covariance matrices. This captures the shape, spread, and orientation of each class distribution.

Method	Before Dimming	Mild	Moderate	Aggressive
Weighted kNN k=15	50.49%	37.01%	34.34%	33.05%
Weighted Mahalanobis	54.24%	52.47%	45.35%	34.02%
Weighted Centroid	35.86%	35.86%	34.86%	34.20%

The plateau at 54% occurs because Mahalanobis operates on a fixed projection. The 256D PCA space captures certain class separations but misses others. The geometry is measured, but the measurement cannot improve the underlying space.

2.2 Evolution Plateau

The freeze-and-stack architecture evolves a projection layer, freezes it, then evolves the next layer on top. Each layer finds useful structure, but without feedback about class geometry, layers can work at cross-purposes. The system converges to local optima where additional layers provide diminishing

returns.

Prior GENREG experiments (v1-v27) demonstrated this: scaling from 4 to 48 seeds improved accuracy from 44.73% to only 52.80%. More of the same approach yields diminishing returns. The breakthrough to 68.20% came from feature diversity, not feature quantity.

3. The Geometric Mahalanobis Cascade

The cascade architecture breaks through both plateaus by interleaving evolution and geometric measurement. Each layer operates on a geometrically transformed space, ensuring the evolutionary search explores new regions of the function space.

3.1 Architecture Overview

Layer 1: Evolve projection P_1 on raw input X
Measure: Compute per-class Mahalanobis geometry M_1
Transform: Apply M_1 to normalize class distributions

Layer 2: Evolve projection P_2 on M_1 -transformed space
Measure: Compute new geometry M_2
Transform: Apply M_2 to reshape space

Layer N: Continue cascade until convergence

3.2 Why Interleaving Works

The key insight is that the Mahalanobis transform acts as a between-layer regularizer. After each evolved projection, the Mahalanobis computation identifies where classes actually live and how they are shaped. The next layer then evolves relative to this measured geometry, not the raw output.

The geometry keeps shifting. Each layer sees a different landscape because the previous Mahalanobis transform rotated and scaled the space according to class covariance. The activations hit different regions of the function space each time. The cascade is not compress → compress → compress. It is project → measure shape → project relative to shape → measure new shape → repeat.

3.3 The Fold/Unfold Connection

This architecture implements the theoretical fold/unfold primitive at the geometric level. Each Mahalanobis transform is a 'fold' - a binary decision about which directions in the current space carry class-discriminative information. The subsequent evolved projection is the 'unfold' - exploring new structure within the folded geometry.

The cascade creates a sequence of binary boundary decisions, each one establishing a new context for the next. This mirrors the theoretical insight that information fundamentally consists of cascaded is/isn't decisions - the same structure that governs protein folding and, more broadly, the boundary between physical reality and information.

4. Mathematical Formulation

4.1 Mahalanobis Distance

For a sample x and class c with centroid μ_c and covariance Σ_c , the Mahalanobis distance is:

$$d_M(x, c) = \sqrt{(x - \mu_c)^T \Sigma_c^{-1} (x - \mu_c)}$$

This distance accounts for the shape of the class distribution. Points along the principal axes of high variance are considered 'closer' than points perpendicular to those axes, even at the same Euclidean distance.

4.2 Geometric Normalization Transform

The Mahalanobis geometry induces a transformation that normalizes class distributions. For each class c , the whitening transform is:

$$W_c = \Sigma_c^{-1/2}$$

In the whitened space, the class covariance becomes identity, and Mahalanobis distance reduces to Euclidean distance. The cascade applies a global version of this transform before each new evolutionary layer.

4.3 Evolved Projection Layer

Each evolved layer consists of random projections with activation functions selected from a catalog of 25,842 non-degenerate curves. The projection is:

$$h = \sigma(Wx + b)$$

where W is evolved via tournament selection, σ is selected from the activation catalog, and fitness is measured by classification accuracy on a validation set.

4.4 Cascade Composition

The full cascade for N layers is:

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X0 = raw input
For i = 1 to N:
  Hi = Pi(Xi-1)    [evolved projection]
  Mi = Mahalanobis(Hi)    [geometric measurement]
  Xi = Whiten(Hi, Mi)    [normalize by geometry]
Output: Classify using MN distances
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5. Theoretical Foundations

5.1 The Infinite Function Space

Prior GENREG experiments optimized within fixed feature vocabularies: HOG, DCT, LBP, and other human-designed representations. These are single points in an infinite space of possible mathematical operations on pixel regions.

Key Realization: Any mathematical function applied to a pixel region creates a unique fingerprint. Some fingerprints happen to align with concepts humans have named (edges, textures, frequencies). But most of the information space is unmapped - functions nobody thought to try because there is no intuitive name for what they extract.

The Geometric Mahalanobis Cascade enables exploration of this infinite space by using geometric feedback to guide the search. Instead of asking 'which known features should I combine?', the system asks 'what mathematical operations create discriminative signal?' with no constraints on what those operations look like.

5.2 The Information Boundary

The architecture is grounded in a deeper theoretical insight about the nature of information. At the most fundamental level, information consists of binary decisions: is/isn't, boundary/not-boundary. This is the 'first rule of information' - the edge where reality meets representation.

Protein folding provides the biological analogy. Proteins do not encode 'how to be an enzyme' - they encode 'fold here, not here', and function emerges from accumulated folding decisions. The information is not in the fold itself but in the sequence of fold/don't-fold choices.

The Mahalanobis transform in each cascade layer implements this principle. It makes a binary determination about each direction in the feature space: does this direction carry class-discriminative information (keep/amplify) or noise (suppress)? The cascade builds structure through accumulated geometric decisions.

5.3 Why Plateaus Break

Both isolated approaches plateau because they operate in a fixed reference frame:

System	Why It Plateaus	What's Missing
Mahalanobis alone	Measures geometry in one projection	Cannot improve the projection
Evolution alone	Optimizes projections blindly	No feedback on class geometry
Interleaved	Neither sees the same space twice	Continuous reshaping breaks plateaus

6. Implementation Strategy

6.1 Algorithm Outline

Input: Training data X , labels Y , max_layers N

1. Initialize:

$X_{\text{current}} = \text{preprocess}(X)$
 $\text{frozen_layers} = []$

2. For layer $i = 1$ to N :

a. Evolve projection P_i on X_{current} using EA
- Population: 60, Tournament selection
- Fitness: kNN accuracy on validation split
- Convergence: stagnation limit

b. Freeze P_i , compute $H_i = P_i(X_{\text{current}})$

c. Compute Mahalanobis geometry M_i :
- Per-class centroids μ_c
- Per-class covariances Σ_c
- Global whitening transform W

d. Transform: $X_{\text{current}} = W @ H_i$

e. Evaluate cascade accuracy
- If no improvement for k layers, stop

3. Output: Stacked frozen layers + final classifier

6.2 Key Design Decisions

Whitening vs. Full Mahalanobis: Use a global whitening transform rather than per-class transforms between layers. This preserves class separation information while normalizing the overall geometry. Per-class transforms can collapse inter-class structure.

Activation Selection: The 25,842-curve catalog provides diversity. Each layer's evolution selects activations suited to the current geometric space, which changes after each Mahalanobis transform.

Dimensionality Management: Each projection can expand or contract dimensionality. The Mahalanobis covariance regularization prevents degenerate dimensions from dominating.

6.3 Expected Behavior

Early layers should find coarse separations (like v21's 58% jump from feature diversity). Middle layers refine boundaries using the normalized geometry. Late layers may show diminishing returns,

signaling convergence. The optimal depth is task-dependent and should be determined empirically.

7. Experimental Roadmap

7.1 Phase 1: Baseline Validation

Implement the basic cascade with 2-3 layers on CIFAR-10. Compare against v27 baseline (68.20%) and isolated Mahalanobis (54.24%). Success criterion: exceed 68.20% with the interleaved approach.

7.2 Phase 2: Depth Exploration

Systematically vary cascade depth from 2 to 10 layers. Measure accuracy vs. depth curve. Identify the point of diminishing returns and characterize the accuracy/compute tradeoff.

7.3 Phase 3: Transform Variations

Experiment with alternative between-layer transforms: partial whitening, class-weighted normalization, learned vs. analytic transforms. Determine which geometric operations provide the best guidance for subsequent evolution.

7.4 Phase 4: Function Space Expansion

Extend beyond the current activation catalog to evolve the mathematical operations themselves. This is the ultimate goal: discovering novel functions that have no human-assigned names but create maximally discriminative structure.

8. Conclusion

The Geometric Mahalanobis Cascade represents a synthesis of two complementary approaches to gradient-free learning. By interleaving evolved projections with geometric class measurement, the architecture escapes the plateau limits that constrain each approach in isolation.

More fundamentally, this architecture begins to implement the theoretical fold/unfold primitive that underlies information itself. Each layer makes binary geometric decisions about which structure matters, building cascaded boundaries that mirror the deep architecture of physical reality meeting representation.

The journey from 38% to 68% on CIFAR-10 without gradients demonstrated that evolutionary pressure can discover discriminative structure. The Geometric Mahalanobis Cascade is the next step: using geometric feedback to guide that evolutionary search through the infinite space of possible mathematical operations.

"The fold primitive isn't a specific function. It's the capacity to apply any function and let selection pressure decide if it matters."